

Twin Commutative Algebra and Algebraic Geometry: Schemes with Dynamic Group Actions and the Geometry of Symmetry

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Abstract

This article develops the algebraic–geometric layer of the twin spaces program. Building on the deformation cohomology theory (Article A), the 2-categorical and operadic framework (Article B), and the rigidity/quantum aspects (Article C), we construct a full-fledged theory of *twin commutative algebra and algebraic geometry*.

A twin ring is a pair (R, G) where R is a commutative ring and G acts on R by automorphisms. From this we derive twin ideals, the twin spectrum, and twin modules. A twin scheme is a scheme equipped with a compatible group action (or group scheme over the base), and twin sheaves are sheaves endowed with equivariant structures. We introduce twin cohomology functors, a twin intersection theory, and a Riemann–Roch theorem for twin varieties. We then construct twin moduli spaces (twin curves, twin Hilbert schemes), and develop applications to public-key cryptography, error-correcting codes, and arithmetic dynamics.

On the arithmetic side, we define twin étale cohomology and formulate a twin version of the Langlands program in which Galois representations and automorphic objects carry an additional twin structure. We conclude with examples (twin Fermat varieties, twin K3 surfaces), computational aspects (twin Gröbner bases and twin cohomology algorithms), and a list of open problems, including connections with physics and motivic integration.

Conceptually, twin algebraic geometry extends Grothendieck’s viewpoint by making symmetries part of the foundational data of schemes, rather than an auxiliary structure added afterward.

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1 Introduction

1.1 Classical background

Classical algebraic geometry studies solutions of polynomial equations via commutative algebra. Affine varieties correspond to prime spectra of rings, projective varieties to homogeneous spectra, and more generally schemes allow arithmetic and geometric information to be handled simultaneously.

The Grothendieck paradigm may be summarized as:

- geometry is encoded by pairs $(X, \mathcal{O}_X)$,
- sheaves and their cohomology carry refined information,
- moduli spaces organize families of geometric objects.

In many natural situations, however, the objects of interest are not just schemes, but schemes with *symmetry*:

- algebraic curves over finite fields come with Frobenius actions,
- complex varieties with real structures carry complex conjugation,
- abelian varieties have rich endomorphism rings and automorphism groups,
- moduli problems often have non-trivial stabilizers and monodromy.

Equivariant algebraic geometry and the theory of group actions on varieties address some of these phenomena, but the symmetry typically appears as an *extra layer* on top of a given scheme.

1.2 The twin extension

The philosophy of twin spaces is that symmetry should be present *from the beginning*: the underlying object and its symmetries form an inseparable pair.

Definition 1 (Twin ring, first informal version). A twin ring is a pair (R, G) where R is a commutative ring with identity and G is a group acting on R by ring automorphisms.

This leads naturally to twin spectra, twin schemes, and twin sheaves.

Remark 2 (Twin perspective). A scheme encodes a topological space together with a sheaf of rings. A twin scheme enhances this by remembering also a group action (possibly varying over the base) and how this action interacts with the local algebra. This is particularly relevant for:

- (a) objects with rich intrinsic symmetry (e.g. modular curves, Shimura varieties),
- (b) families with monodromy or varying symmetries,
- (c) situations where symmetry-breaking and deformations are essential (physics, dynamics).

Example 3 (Natural sources of twin structure). Typical examples include:

- algebraic curves $C/\mathbb{F}_q$ with Frobenius endomorphism and $\text{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$,
- varieties over $\mathbb{R}$ with complex conjugation action of $\mathbb{Z}/2\mathbb{Z}$,
- abelian varieties with an action of an endomorphism ring or of a finite group,
- Calabi–Yau varieties with symmetry groups relevant in string theory.

1.3 Position in the twin spaces program

This article assumes the deformation and cohomological framework of twin operations developed in Article A, the 2-categorical and operadic structure of twin spaces in Article B, and the rigidity/quantum aspects of Article C. The present work is the *algebraic geometry layer* of the program.

Two features distinguish this layer from classical equivariant geometry. First, the invariant subring R^G and the invariant cohomology $H^i(X, \mathcal{F})^G$ are not afterthoughts but the primary global invariants of a twin object; when $|G|$ is invertible they are exact functors (Lemma 28) that behave like ordinary cohomology, whereas in the modular case they must be derived. Second, numerical invariants such as the Euler characteristic become *averages over the group*, as the twin Riemann–Roch theorem (Theorem 42) makes precise. The slogan throughout is that a twin scheme is a scheme whose symmetry is part of its definition, so every classical construction is replaced by its G -equivariant, then G -invariant, counterpart.

The plan is as follows. In Section 2 we define twin rings and modules. In Section 3 we introduce twin schemes and the twin spectrum functor. In Section 4 we develop twin sheaves and twin (co)homology. In Section 5 we set up twin intersection theory and twin Chow groups. In Section 6 we prove a twin Riemann–Roch theorem. In Section 7 we construct twin moduli spaces. In Section 8 we discuss cryptographic applications. In Section 9 we define twin étale cohomology. In Section 10 we sketch a twin Langlands picture. In Section 11 we address computational issues. In Section 12 we present examples. We conclude in Section 13 with open problems and connections to physics.

2 Twin Rings and Twin Modules

2.1 Basic definitions

Definition 4 (Twin ring). A *twin ring* is a pair (R, G) where:

- R is a commutative ring with 1,
- G is a group,
- G acts on R by ring automorphisms

$$\rho : G \times R \rightarrow R, \quad (g, r) \mapsto g \cdot r,$$

such that

$$g \cdot (r + s) = g \cdot r + g \cdot s, \quad g \cdot (rs) = (g \cdot r)(g \cdot s), \quad g \cdot 1 = 1.$$

We write R^G for the invariant subring, i.e. $R^G = \{r \in R : g \cdot r = r, \forall g \in G\}$.

Example 5 (Symmetric polynomials). Let $R = k[x_1, \dots, x_n]$ and $G = S_n$ acting by permutation of variables. Then (R, S_n) is a twin ring and R^{S_n} is the ring of symmetric polynomials. By the fundamental theorem of symmetric polynomials, $R^{S_n} = k[e_1, \dots, e_n]$ is itself a polynomial ring in the elementary symmetric functions, so $\text{Spec } R^{S_n} \cong \mathbb{A}^n$ and the quotient map $\mathbb{A}^n = \text{Spec } R \rightarrow \text{Spec } R^{S_n} = \mathbb{A}^n$, $(x_1, \dots, x_n) \mapsto (e_1, \dots, e_n)$, is finite flat of degree $n!$, branched over the discriminant locus. This is the prototype twin spectrum: the geometry of the quotient encodes precisely the S_n -symmetry.

Example 6 (Cyclotomic extension). Let $R = \mathbb{Z}[\zeta_n]$ where ζ_n is a primitive n -th root of unity, and let $G = (\mathbb{Z}/n\mathbb{Z})^\times$ act by $a \cdot \zeta_n = \zeta_n^a$. Then (R, G) is a twin ring; $R^G = \mathbb{Z}$ reflects the Galois structure of the cyclotomic field.

2.2 Twin modules

Definition 7 (Twin module). Let (R, G) be a twin ring. A *twin R -module* is an R -module M together with an action of G by R -linear automorphisms

$$G \times M \rightarrow M, \quad (g, m) \mapsto g \cdot m$$

such that

$$g \cdot (rm) = (g \cdot r)(g \cdot m),$$

for all $g \in G, r \in R, m \in M$.

Remark 8. Twin modules are precisely $R \rtimes G$ -modules under the usual skew group ring construction. This is the algebraic shadow of the twin operations viewpoint of Articles A and B.

2.3 The invariant subring and twin ideals

Definition 9 (Invariant subring and action kernel). Let (R, G) be a twin ring. The *invariant subring* is

$$R^G := \{ r \in R : g \cdot r = r \text{ for all } g \in G \}.$$

It is a subring of R containing 1; in general it is *not* an ideal of R (an ideal containing 1 would be all of R). The *kernel* of the action is the normal subgroup

$$N := \{ g \in G : g \cdot r = r \text{ for all } r \in R \} \trianglelefteq G.$$

Proposition 10. R^G is a subring of R , and the action of G on R factors through a faithful action of the quotient group G/N ; in particular $N \neq G$ whenever the action is non-trivial. The inclusion $R^G \hookrightarrow R$ induces the quotient morphism $\text{Spec } R \rightarrow \text{Spec } R^G$.

Proof. R^G is closed under addition and multiplication and contains 1, hence is a subring. (It is not an ideal: for $r \in R^G$ and $s \in R$ one has $g \cdot (sr) = (g \cdot s)r$, which differs from sr as soon as $g \cdot s \neq s$, so $sr \notin R^G$ in general.) The assignment $g \mapsto (r \mapsto g \cdot r)$ is a homomorphism $G \rightarrow \text{Aut}(R)$ with kernel exactly N ; hence G/N acts faithfully, and $N \neq G$ precisely when the action is non-trivial. $\square$

Definition 11 (Twin ideal). An ideal $I \subset R$ is called a *twin ideal* if it is stable under G :

$$g \cdot I \subseteq I \quad \text{for all } g \in G.$$

If equality holds for all g , we call I *G -invariant*.

Remark 12. Twin ideals are the natural objects to consider for defining sub-twin rings, quotients, and spectra with symmetry. They form a lattice stable under finite intersections and sums.

2.4 Prime twin ideals and orbit structure

Definition 13 (Twin prime). A proper *prime* ideal $P \subset R$ is called *twin-prime* if:

- P is a prime ideal of R ,
- its G -orbit $\{g \cdot P : g \in G\}$ is finite.

Remark 14. The finiteness of the orbit ensures that the corresponding points in the spectrum form a controlled configuration under the group action. For infinite groups one often restricts attention to orbits of finite type or closed orbits.

Proposition 15. *Let (R, G) be Noetherian. Then:*

- (a) *the set of twin-prime ideals is stable under the G -action,*
- (b) *every twin ideal can be written as an intersection of twin-prime ideals (possibly after saturation).*

Idea of proof. For (a), stability under G is immediate. For (b), use the classical primary decomposition for G -invariant ideals and group the components along G -orbits, taking intersections of the conjugates. Details follow standard arguments in equivariant algebra with the twin constraints built in. $\square$

3 Twin Schemes

3.1 Affine twin schemes

Definition 16 (Affine twin scheme). Let (R, G) be a twin ring. The *affine twin scheme* associated to (R, G) is the pair

$$(X, G) := (\operatorname{Spec} R, G),$$

where G acts on $\operatorname{Spec} R$ by

$$g \cdot \mathfrak{p} := \{g \cdot r : r \in \mathfrak{p}\}$$

and on the structure sheaf $\mathcal{O}_{\operatorname{Spec} R}$ via the induced action on localizations.

Remark 17. The action on the structure sheaf is compatible with restriction and localization, so we obtain an action by automorphisms of ringed spaces. Affine twin schemes are thus objects of the 2-category of ringed spaces with group actions.

3.2 Twin spectrum functor

Definition 18 (Twin spectrum). The *twin spectrum* functor

$$\operatorname{Spec}^{\text{tw}} : (\text{Twin rings})^{\text{op}} \rightarrow \text{Twin schemes}$$

sends (R, G) to $(\operatorname{Spec} R, G)$ and a morphism of twin rings

$$\phi : (R, G) \rightarrow (S, H)$$

(consisting of a ring map $\phi_R : R \rightarrow S$ and a group homomorphism $\phi_G : G \rightarrow H$ compatible with the actions) to the twin morphism

$$(\operatorname{Spec} \phi_R, \phi_G^*) : (\operatorname{Spec} S, H) \rightarrow (\operatorname{Spec} R, G),$$

where $\operatorname{Spec} \phi_R$ is the usual map of schemes and ϕ_G^* is the induced action on group data.

Proposition 19. *$\operatorname{Spec}^{\text{tw}}$ is functorial, and restricts to the usual spectrum functor on the underlying rings.*

Proof. Functoriality is checked componentwise on rings and groups. Compatibility of the group action with the inverse image of primes ensures that the twin structure is preserved. $\square$

3.3 Gluing twin affine pieces

Definition 20 (Twin scheme). A *twin scheme* is a pair (X, G) where:

- X is a scheme,
- G is a group (or group scheme) acting on X by automorphisms,
- there exists an open affine cover $\{U_i\}$ of X such that each (U_i, G) is isomorphic to an affine twin scheme $(\text{Spec } R_i, G)$ for some twin ring (R_i, G) , and the gluing maps are G -equivariant.

Proposition 21. *Twin schemes and twin morphisms (scheme morphisms compatible with the group actions) form a category, and affine twin schemes are precisely the objects isomorphic to $\text{Spec}^{\text{tw}}(R, G)$.*

Proof. Gluing along G -equivariant isomorphisms is standard in descent theory for schemes. The compatibility conditions ensure that the resulting action is globally defined and the structure sheaf is preserved. $\square$

Example 22 (Twin projective space). Let $R = k[x_0, \dots, x_n]$ with $G \subset \text{Aut}_k(R)$ a group of homogeneous automorphisms. Then $X = \text{Proj } R$ carries an induced G -action, and $(\text{Proj } R, G)$ is a twin scheme.

4 Twin Sheaves and Twin Cohomology

4.1 Twin sheaves

Definition 23 (Twin $\mathcal{O}_X$ -module). Let (X, G) be a twin scheme. A *twin sheaf* (or twin $\mathcal{O}_X$ -module) on (X, G) is a sheaf of $\mathcal{O}_X$ -modules $\mathcal{F}$ together with, for each $g \in G$, an isomorphism of sheaves

$$\theta_g : g^* \mathcal{F} \xrightarrow{\sim} \mathcal{F}$$

such that $\theta_1 = \text{id}$ and $\theta_{gh} = \theta_h \circ h^* \theta_g$ for all $g, h \in G$.

Remark 24. This is the usual notion of a G -equivariant sheaf, but in the twin viewpoint it is part of the intrinsic structure: twin schemes are always considered together with their twin sheaves.

Definition 25 (Twin quasi-coherent and coherent sheaves). A twin sheaf $\mathcal{F}$ is called *twin quasi-coherent* (resp. *twin coherent*) if the underlying $\mathcal{O}_X$ -module is quasi-coherent (resp. coherent).

4.2 Twin cohomology

Definition 26 (Twin cohomology). Let $\mathcal{F}$ be a twin sheaf on (X, G) . The *twin cohomology* of $\mathcal{F}$ is defined as

$$H_{\text{tw}}^i(X, \mathcal{F}) := H^i(X, \mathcal{F})^G,$$

the G -invariant part of the usual sheaf cohomology.

Remark 27. One can refine this by considering the full representation $H^i(X, \mathcal{F})$ as a G -module and analyzing its decomposition, but for the present article we focus on the invariant part, which is the natural receptacle for global twin structures.

Lemma 28 (Averaging and exactness of invariants). *Suppose G is finite and $|G|$ is invertible in the base field (e.g. characteristic zero). Then the Reynolds operator $e = \frac{1}{|G|} \sum_{g \in G} g$ is an idempotent endomorphism of every G -module M with image M^G , giving a canonical splitting $M = M^G \oplus (1-e)M$. Hence the functor $(-)^G$ is exact, and $H_{\text{tw}}^i(X, \mathcal{F}) = H^i(X, \mathcal{F})^G$ is a direct summand of $H^i(X, \mathcal{F})$ on which twin cohomology inherits the long exact sequence of any short exact sequence of twin sheaves. In the modular case ($\text{char} \mid |G|$) exactness fails and the correct object is the G -hypercohomology, computed by a spectral sequence $H^p(G, H^q(X, \mathcal{F})) \Rightarrow \mathbb{H}^{p+q}$.*

Proof. Idempotency $e^2 = e$ and $eM = M^G$ are the standard properties of the Reynolds operator; $1 - e$ projects onto a G -stable complement. A functor that is a direct-summand projection is exact, and exact functors commute with the formation of cohomology of complexes, giving the stated long exact sequence. The modular statement is the Hochschild–Serre/equivariant hypercohomology spectral sequence. $\square$

Proposition 29 (Functoriality). *Twin cohomology is functorial in the pair $(X, \mathcal{F})$ and compatible with pullback and pushforward along twin morphisms under standard hypotheses (properness, flatness, etc.).*

Idea. Functoriality of usual cohomology together with the equivariance of morphisms of twin schemes and twin sheaves implies that the induced maps preserve G -invariants. $\square$

4.3 Twin Čech cohomology

Let $\mathfrak{U} = \{U_i\}$ be a G -stable open cover of X , i.e. gU_i is a union of U_j ’s for all $g \in G$.

Definition 30 (Twin Čech complex). The *twin Čech complex* associated with $\mathfrak{U}$ and a twin sheaf $\mathcal{F}$ is the usual Čech complex

$$\check{C}^\bullet(\mathfrak{U}, \mathcal{F}),$$

viewed as a complex of G -modules via the twin structure. Its G -invariant subcomplex

$$\check{C}_{\text{tw}}^\bullet(\mathfrak{U}, \mathcal{F}) := \check{C}^\bullet(\mathfrak{U}, \mathcal{F})^G$$

computes the twin Čech cohomology

$$\check{H}_{\text{tw}}^i(\mathfrak{U}, \mathcal{F}) := H^i(\check{C}_{\text{tw}}^\bullet(\mathfrak{U}, \mathcal{F})).$$

Theorem 31 (Comparison). *If $\mathcal{F}$ is twin quasi-coherent on a separated, quasi-compact twin scheme and $\mathfrak{U}$ is a sufficiently fine G -stable affine cover, then*

$$\check{H}_{\text{tw}}^i(\mathfrak{U}, \mathcal{F}) \cong H_{\text{tw}}^i(X, \mathcal{F}).$$

Sketch. A G -stable affine cover computes $H^i(X, \mathcal{F})$ because X is separated and quasi-compact and $\mathcal{F}$ is quasi-coherent. By the averaging Lemma 28, when $|G|$ is invertible the exact functor $(-)^G$ commutes with the cohomology of the Čech complex, so $\check{H}_{\text{tw}}^i(\mathfrak{U}, \mathcal{F}) = \check{H}^i(\mathfrak{U}, \mathcal{F})^G \cong H^i(X, \mathcal{F})^G = H_{\text{tw}}^i(X, \mathcal{F})$. In the modular case the same holds after replacing invariants by G -hypercohomology. $\square$

5 Twin Intersection Theory

5.1 Twin divisors

Let (X, G) be an integral twin scheme, normal and separated of finite type.

Definition 32 (Twin Weil divisor). A *twin Weil divisor* on (X, G) is a finite formal $\mathbb{Z}$ -linear combination of irreducible codimension-one subvarieties of X that are G -stable, i.e.

$$D = \sum_i n_i Z_i, \quad n_i \in \mathbb{Z},$$

with each Z_i invariant under the action of G . We write $\text{Div}_{\text{tw}}(X)$ for the group of twin Weil divisors.

Definition 33 (Twin Cartier divisor). A *twin Cartier divisor* is a Cartier divisor D on X represented by local equations compatible with the G -action, i.e. for each $g \in G$ the pullback g^*D is linearly equivalent to D . We denote by $\text{Pic}_{\text{tw}}(X)$ the group of isomorphism classes of twin line bundles.

Remark 34. The twin Picard group is a subgroup of the usual Picard group $\text{Pic}(X)$, consisting of line bundles endowed with a G -equivariant structure.

5.2 Twin Chow groups

Definition 35 (Twin cycles). For $k \geq 0$, a *twin k -cycle* on (X, G) is a formal linear combination (with integer coefficients) of irreducible closed subschemes of dimension k that are stabilized by G . The group of twin k -cycles is denoted $Z_k^{\text{tw}}(X)$.

Definition 36 (Twin rational equivalence). Two twin cycles are said to be twin-rationally equivalent if their difference is a finite sum of cycles of the form $\text{div}(f)$ where f is a rational function compatible with the G -action. The quotient group

$$\text{CH}_k^{\text{tw}}(X) := Z_k^{\text{tw}}(X) / \sim_{\text{tw}}$$

is called the *twin Chow group* of dimension k .

Remark 37. Equivalently, one can define $\text{CH}_k^{\text{tw}}(X)$ as the G -invariant part of the usual Chow group, endowed with the induced intersection product.

5.3 Intersection pairings

Under usual smoothness and properness assumptions, $\text{CH}_*^{\text{tw}}(X)$ carries an intersection product

$$\cdot : \text{CH}_i^{\text{tw}}(X) \times \text{CH}_j^{\text{tw}}(X) \rightarrow \text{CH}_{i+j-d}^{\text{tw}}(X),$$

where $d = \dim X$.

Proposition 38. *The intersection product of cycles descends to twin Chow groups and is compatible with the G -action. In particular, the G -invariant part of the classical intersection pairing induces a well-defined pairing on $\text{CH}_*^{\text{tw}}(X)$.*

Idea. Since the intersection product is functorial and G acts by automorphisms of the scheme, $g \cdot (Z \cdot Z') = (g \cdot Z) \cdot (g \cdot Z')$ for all cycles. Passing to G -invariant cycles and rational equivalence classes yields the result. $\square$

6 Twin Riemann–Roch Theorem

We now outline a Riemann–Roch formula in the twin setting.

6.1 Twin Chern and Todd classes

Let (X, G) be a smooth projective twin scheme over a field of characteristic zero, and let $\mathcal{E}$ be a twin coherent sheaf (e.g. a twin vector bundle) on X .

Definition 39 (Twin Chern character). The *twin Chern character* of $\mathcal{E}$ is defined as

$$\mathrm{ch}_{\mathrm{tw}}(\mathcal{E}) := \mathrm{ch}(\mathcal{E})^G \in \bigoplus_i \mathrm{CH}_{\mathrm{tw}}^i(X) \otimes \mathbb{Q},$$

the G -invariant part of the usual Chern character.

Definition 40 (Twin Todd class). The *twin Todd class* of X is

$$\mathrm{Td}_{\mathrm{tw}}(X) := \mathrm{Td}(T_X)^G \in \bigoplus_i \mathrm{CH}_{\mathrm{tw}}^i(X) \otimes \mathbb{Q}.$$

6.2 The twin Riemann–Roch statement

Definition 41 (Twin Euler characteristic). The *twin Euler characteristic* of $\mathcal{E}$ is

$$\chi_{\mathrm{tw}}(X, \mathcal{E}) := \sum_{i \geq 0} (-1)^i \dim H_{\mathrm{tw}}^i(X, \mathcal{E}).$$

Theorem 42 (Twin Grothendieck–Riemann–Roch). *Let (X, G) be a smooth projective twin scheme and $\mathcal{E}$ a twin coherent sheaf. Then*

$$\chi_{\mathrm{tw}}(X, \mathcal{E}) = \frac{1}{|G|} \sum_{g \in G} \chi(g, \mathcal{E}), \quad \chi(g, \mathcal{E}) := \sum_{i \geq 0} (-1)^i \mathrm{tr}(g \mid H^i(X, \mathcal{E})),$$

where $\chi(g, \mathcal{E})$ is the g -equivariant Euler characteristic, computed by the holomorphic Lefschetz fixed-point formula of Atiyah–Bott as an integral over the fixed locus X^g .

Idea of proof. By the averaging Lemma 28, for finite G in characteristic zero $\dim H^i(X, \mathcal{E})^G = \frac{1}{|G|} \sum_g \mathrm{tr}(g \mid H^i(X, \mathcal{E}))$, the rank of the Reynolds idempotent. Summing with signs,

$$\chi_{\mathrm{tw}}(X, \mathcal{E}) = \sum_i (-1)^i \dim H^i(X, \mathcal{E})^G = \frac{1}{|G|} \sum_{g \in G} \sum_i (-1)^i \mathrm{tr}(g \mid H^i) = \frac{1}{|G|} \sum_{g \in G} \chi(g, \mathcal{E}).$$

Each $\chi(g, \mathcal{E})$ is given by the holomorphic Lefschetz (Atiyah–Bott) fixed-point theorem as an integral over X^g ; the term $g = e$ contributes $\chi(e, \mathcal{E}) = \chi(X, \mathcal{E}) = \int_X \mathrm{ch}(\mathcal{E}) \mathrm{Td}(X)$ by ordinary Grothendieck–Riemann–Roch. This proves the averaged formula. $\square$

Remark 43. In the presence of non-trivial stabilizers, local contributions from fixed-point loci of the G -action appear, leading to refined formulas reminiscent of the Atiyah–Segal–Singer index theorem in the equivariant setting.

Remark 44 (The naive product formula is *not* twin Riemann–Roch). One might guess $\chi_{\mathrm{tw}}(X, \mathcal{E}) = \int_X \mathrm{ch}_{\mathrm{tw}}(\mathcal{E}) \cdot \mathrm{Td}_{\mathrm{tw}}(X)$ with the invariant characteristic classes defined above. This is false in general, because the invariant part of a product is not the product of invariant parts. Take $X = \mathbb{P}^1$, $G = \mathbb{Z}/2$ acting by $[x_0 : x_1] \mapsto [x_0 : -x_1]$, and $\mathcal{E} = \mathcal{O}(1)$: then $H^0 = \langle x_0, x_1 \rangle$ has σ -eigenvalues ± 1 , so $\dim(H^0)^G = 1$ and $\chi_{\mathrm{tw}} = 1$; but G acts trivially on $\mathrm{CH}(\mathbb{P}^1)$, whence $\mathrm{ch}_{\mathrm{tw}} = 1 + [\mathrm{pt}]$, $\mathrm{Td}_{\mathrm{tw}} = 1 + [\mathrm{pt}]$, and $\int_{\mathbb{P}^1} (1 + [\mathrm{pt}])^2 = 2 \neq 1$. The averaged formula gives the correct value $\frac{1}{2}(\chi(e) + \chi(\sigma)) = \frac{1}{2}(2 + 0) = 1$. The product formula holds only when G acts trivially on every $H^i(X, \mathcal{E})$.

7 Twin Moduli Spaces

7.1 Moduli of twin curves

Let (C, G) be a smooth projective twin curve of genus $g \geq 2$ over an algebraically closed field k , where G acts by automorphisms.

Definition 45 (Twin curve). A *twin curve* of type (g, G) is a pair (C, G) where C is a smooth projective curve of genus g and $G \subset \text{Aut}(C)$ is a fixed finite group acting faithfully on C .

Definition 46 (Moduli functor). The moduli functor of twin curves assigns to a k -scheme S the groupoid of families

$$\pi : \mathcal{C} \rightarrow S$$

of smooth proper curves of genus g together with an action of the constant group scheme G_S on $\mathcal{C}$ over S , satisfying natural stability conditions.

Theorem 47 (Existence of moduli). *Under standard stable reduction hypotheses, the moduli functor of twin curves of type (g, G) is represented (or at least corepresented) by a Deligne–Mumford stack $\mathcal{M}_g^{\text{tw}}(G)$ equipped with a natural twin structure.*

Remark 48. The stack $\mathcal{M}_g^{\text{tw}}(G)$ sits inside the usual moduli stack of curves as a closed substack parameterizing curves with specified automorphism group data. The twin viewpoint promotes this to the *primary* moduli object, not just a special locus.

7.2 Twin Hilbert schemes

Definition 49 (Twin Hilbert functor). Given a twin scheme (X, G) and a fixed Hilbert polynomial P , the *twin Hilbert functor* sends a k -scheme S to the groupoid of closed subschemes $Z \subset X \times S$ flat over S with Hilbert polynomial P and stable under the action of G (i.e. G acts fiberwise on Z).

Theorem 50 (Twin Hilbert scheme). *Under suitable Noetherian hypotheses, the twin Hilbert functor is represented by a twin scheme $\text{Hilb}_P^{\text{tw}}(X, G)$, called the twin Hilbert scheme.*

Idea. One constructs the usual Hilbert scheme and then considers the closed subscheme where the universal family is G -stable. The action of G on X induces an action on the Hilbert scheme, and the fixed locus captures twin structures. $\square$

8 Applications to Cryptography and Coding Theory

8.1 The hidden kernel problem

Twin rings and twin schemes give rise to new hardness assumptions for cryptography.

Definition 51 (Hidden kernel problem (HKP)). Let (R, G) be a twin ring and $\varphi : R \rightarrow S$ a surjective ring homomorphism to another ring S carrying a compatible G -action. The *hidden kernel problem* asks, given black-box access to φ and to the G -action, to recover the kernel ideal $\text{Ker } \varphi$ (or its G -saturation).

Remark 52. The twin structure makes the kernel invariant under G and often highly non-trivial combinatorially, suggesting potential post-quantum hardness when G is large or when the geometry of $\text{Spec } R$ is complicated.

8.2 Twin elliptic curves

Example 53 (Twin elliptic curves). Let E/k be an elliptic curve and G a finite subgroup of $\text{Aut}(E)$ (for example, complex multiplication or Frobenius over finite fields). Then (E, G) is a twin curve, and one can define twin endomorphism rings, twin isogenies, and twin torsion points. These structures yield variants of isogeny-based cryptosystems where the extra symmetry may be used either as a feature (extra structure for protocols) or as a parameter to hide.

8.3 Twin algebraic codes

Definition 54 (Twin algebraic–geometry code). Let (X, G) be a twin curve over a finite field, $\mathcal{L}$ a twin line bundle, and $P_1, \dots, P_n \in X(k)$ a collection of rational points forming G -orbits. The associated *twin algebraic–geometry code* is defined by evaluating global twin sections

$$H_{\text{tw}}^0(X, \mathcal{L}) \subset H^0(X, \mathcal{L})$$

at the points P_i .

Remark 55. Twin codes inherit symmetry properties from G that can be exploited to design codes with large automorphism groups, potentially useful for code-based cryptography and efficient decoding algorithms.

9 Twin Étale Cohomology

9.1 Twin étale site

Let (X, G) be a twin scheme over a base where étale cohomology is well-behaved (e.g. Noetherian, finite type).

Definition 56 (Twin étale site). The *twin étale site* $X_{\text{ét}}^{\text{tw}}$ is the usual étale site of X equipped with the induced G -action on objects and morphisms. A *twin étale sheaf* is an étale sheaf together with a compatible G -equivariant structure.

Definition 57 (Twin étale cohomology). For a twin étale sheaf $\mathcal{F}$ on $X_{\text{ét}}^{\text{tw}}$, the *twin étale cohomology* is

$$H_{\text{ét}, \text{tw}}^i(X, \mathcal{F}) := H_{\text{ét}}^i(X, \mathcal{F})^G.$$

Remark 58. For constant coefficients $\mathcal{F} = \underline{A}$ with a non-trivial G -action, this captures simultaneously arithmetic information (via the étale site) and symmetry information.

9.2 ℓ -adic twin cohomology

Let ℓ be a prime different from the residue characteristic. Take an inverse system of twin étale sheaves $(\mathcal{F}_n)_{n \geq 1}$ with $\mathcal{F}_n$ ℓ^n -torsion and transition maps compatible with G .

Definition 59 (ℓ -adic twin cohomology). The ℓ -adic *twin cohomology* is defined as

$$H_{\text{ét}, \text{tw}}^i(X, \mathbb{Z}_\ell) := \varprojlim_n H_{\text{ét}}^i(X, \mathcal{F}_n)^G.$$

Remark 60. The resulting $\mathbb{Z}_\ell$ -module carries an action of the absolute Galois group and of G , and thus is a natural receptacle for twin Galois representations.

10 The Twin Langlands Program

10.1 Twin Galois representations

Let k be a global field and X/k a smooth projective twin scheme.

Definition 61 (Twin Galois representation). An ℓ -adic twin Galois representation is a continuous representation

$$\rho_{\text{tw}} : \text{Gal}(\bar{k}/k) \rightarrow \text{Aut}_{\mathbb{Z}_\ell}(H_{\text{ét,tw}}^i(X_{\bar{k}}, \mathbb{Z}_\ell))$$

compatible with the G -action.

Remark 62. Classical ℓ -adic representations are recovered by forgetting the G -structure. The twin data encode additional symmetry coming from the twin scheme.

10.2 Twin automorphic forms

Definition 63 (Twin automorphic object, very rough). A *twin automorphic object* is an automorphic function, form, or representation on an adelic group equipped with a compatible action of a group G of symmetries, arising from the twin geometry of a twin scheme or a twin moduli space.

Conjecture 64 (Twin Langlands correspondence, schematic). *There exists a correspondence between suitable classes of twin Galois representations and twin automorphic objects, intertwining:*

- the action of $\text{Gal}(\bar{k}/k)$ on twin étale cohomology;
- the Hecke action and the twin symmetry G on automorphic side.

Remark 65. This is a long-term direction. Twin structures could provide a cleaner way to package internal symmetries of varieties into the Langlands formalism, especially for moduli spaces with non-trivial automorphism groups.

11 Computational Aspects

11.1 Gröbner bases for twin ideals

Definition 66 (Twin Gröbner basis). Let (R, G) be a twin polynomial ring over a field, and $I \subset R$ a twin ideal. A *twin Gröbner basis* of I is a Gröbner basis G_I that is stable under G : $g \cdot G_I = G_I$ for all $g \in G$.

Remark 67. Twin Gröbner bases reflect both algebraic and symmetry constraints and can drastically reduce redundancy in computations, since one can work up to the G -action.

11.2 Computing twin cohomology

For twin sheaves on projective twin schemes, one can adapt standard algorithms for sheaf cohomology (complexes from free resolutions, Čech complexes) and then take G -invariants at the cochain level.

Proposition 68 (Algorithmic outline). *Assuming effective presentations of R , G , and $\mathcal{F}$, one can compute $H_{\text{tw}}^i(X, \mathcal{F})$ by:*

- (i) constructing a cochain complex computing $H^i(X, \mathcal{F})$,

(ii) describing the G -action on each degree,

(iii) taking invariants degreewise and computing cohomology of the subcomplex.

Remark 69. Practical implementations require careful control of complexity, but the presence of symmetry often reduces the size of the problem significantly.

12 Examples

12.1 Twin Fermat varieties

Example 70 (Twin Fermat hypersurface). Let $X \subset \mathbb{P}^{n+1}$ be the Fermat hypersurface defined by

$$x_0^d + \cdots + x_{n+1}^d = 0$$

over a field containing the d -th roots of unity, and let G be the group of diagonal symmetries generated by multiplying coordinates by roots of unity with product 1. Then (X, G) is a twin variety with rich symmetry. Twin cohomology isolates the G -invariant part of the Hodge structure and is expected to relate to special values of L -functions and periods.

12.2 Twin K3 surfaces

Example 71 (Twin K3). Let X be a K3 surface with a finite symplectic automorphism group G . Then (X, G) is a twin K3 surface. Twin intersection theory on $\mathrm{CH}_{\mathrm{tw}}^1(X)$ captures the G -fixed part of the Néron–Severi group, and twin étale cohomology provides refined Galois representations with extra symmetry. Such objects are natural test beds for the twin Langlands conjectures.

13 Future Directions and Connections to Physics

We conclude with a brief list of directions suggested by the preceding theory.

13.1 Open problems

- **Twin resolution and minimal models.** Develop a theory of minimal models in the twin category, compatible with G -actions and twin cohomology.
- **Twin motives.** Construct a twin version of the category of motives where correspondences carry twin structure.
- **Twin Donaldson–Thomas invariants.** Define enumerative invariants for twin Calabi–Yau threefolds counting twin-stable objects.
- **Effective algorithms.** Provide complexity bounds for computing twin Hilbert schemes and twin cohomology for explicit examples.

13.2 Connections to physics

Twin schemes appear naturally in contexts where symmetry plays a central rôle:

- *String theory and mirror symmetry.* Calabi–Yau manifolds with large symmetry groups suggest studying twin versions of mirror symmetry.
- *Gauge theory.* Moduli of bundles or sheaves with structure group G over a twin base could reveal new dualities.

- *Quantum field theory*. Twin operations and quantum twin spaces (Article C) might be related to algebraic structures underlying topological quantum field theories.

These perspectives reinforce the idea that twin algebraic geometry is not only a natural extension of classical scheme theory, but also a bridge between algebraic geometry, representation theory, and mathematical physics.

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